

FSUJ2830

## Buffer solution pH 5 (phthalate) Traceable to NIST

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**产品说明:** 缓冲水溶液pH 5 (邻苯二甲酸盐) (2.04%邻苯二甲酸氢钾, 0.37%氯化钾, 0.2%氢氧化钠)  
**Product Description:** Buffer solution pH 5 (phthalate) Traceable to NIST

**Cat No. :** J/2830/15, J/2830/17, J/2830/08

**Supplier**

**UK entity/business name**  
Fisher Scientific UK  
Bishop Meadow Road, Loughborough,  
Leicestershire LE11 5RG, United Kingdom

**EU entity/business name**  
Thermo Fisher Scientific  
Janssen Pharmaceuticaaan 3a  
2440 Geel, Belgium

**Emergency Telephone Number** Tel: 01509 231166  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001-703-527-3887

**E-mail address** begel.sdsdesk@thermofisher.com

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### SECTION 2. HAZARD IDENTIFICATION

**Physical State**  
Liquid

**Appearance**  
Colorless

**Odor**  
No information available

#### Emergency Overview

The product contains no substances which at their given concentration are considered to be hazardous to health.

#### Classification of the substance or mixture

Based on available data, the classification criteria are not met

#### Label Elements

None required

#### Physical and Chemical Hazards

None identified.

#### Health Hazards

The product contains no substances which at their given concentration are considered to be hazardous to health.

#### Environmental hazards

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants. . Will likely be mobile in the environment due to its water solubility. The product is water soluble, and may spread in water systems.

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### Other Hazards

This product does not contain any known or suspected endocrine disruptors.

## SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                        | CAS No    | Weight % |
|--------------------------------------------------|-----------|----------|
| Water                                            | 7732-18-5 | > 95     |
| 1,2-Benzenedicarboxylic acid, monopotassium salt | 877-24-7  | < 5      |
| Potassium chloride                               | 7447-40-7 | < 0.5    |
| Sodium hydroxide                                 | 1310-73-2 | < 0.5    |

## SECTION 4. FIRST AID MEASURES

### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

### Inhalation

Remove to fresh air. If breathing is difficult, give oxygen. Get medical attention immediately if symptoms occur.

### Ingestion

Do NOT induce vomiting. Get medical attention.

### Most important symptoms and effects

No information available.

### Self-Protection of the First Aider

No special precautions required.

### Notes to Physician

Treat symptomatically.

## SECTION 5. FIRE-FIGHTING MEASURES

### Suitable Extinguishing Media

Substance is nonflammable; use agent most appropriate to extinguish surrounding fire.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6. ACCIDENTAL RELEASE MEASURES

### Personal Precautions

Ensure adequate ventilation. Use personal protective equipment as required.

### Environmental Precautions

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Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7. HANDLING AND STORAGE

### Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Storage

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific Use(s)

Use in laboratories

## SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component        | China                        | Taiwan                   | Thailand                 | Hong Kong                    |
|------------------|------------------------------|--------------------------|--------------------------|------------------------------|
| Sodium hydroxide | Ceiling: 2 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> | TWA: 2 mg/m <sup>3</sup> | Ceiling: 2 mg/m <sup>3</sup> |

| Component        | ACGIH TLV                    | OSHA PEL                                                 | NIOSH                                                      | The United Kingdom       | European Union |
|------------------|------------------------------|----------------------------------------------------------|------------------------------------------------------------|--------------------------|----------------|
| Sodium hydroxide | Ceiling: 2 mg/m <sup>3</sup> | Ceiling: 2 mg/m <sup>3</sup><br>TWA: 2 mg/m <sup>3</sup> | IDLH: 10 mg/m <sup>3</sup><br>Ceiling: 2 mg/m <sup>3</sup> | 2 mg/m <sup>3</sup> STEL |                |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

#### Personal protective equipment

##### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

##### Hand Protection

Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard | Glove comments        |
|----------------|-------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers | -               | EN 374      | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |             |                       |
| Neoprene       |                   |                 |             |                       |
| PVC            |                   |                 |             |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

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|                                        |                                                                                                                                                                                                        |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | No protective equipment is needed under normal use conditions.                                                                                                                                         |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Particle filter |
| <b>Small scale/Laboratory use</b>      | Maintain adequate ventilation                                                                                                                                                                          |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                 |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                              |

## SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                                  |                          |                                          |
|--------------------------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>                                | Colorless                |                                          |
| <b>Physical State</b>                            | Liquid                   |                                          |
| <b>Odor</b>                                      | No information available |                                          |
| <b>Odor Threshold</b>                            | No data available        |                                          |
| <b>pH</b>                                        | 5                        |                                          |
| <b>Melting Point/Range</b>                       | No data available        |                                          |
| <b>Softening Point</b>                           | No data available        |                                          |
| <b>Boiling Point/Range</b>                       | No information available |                                          |
| <b>Flash Point</b>                               | No data available        | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                          | No data available        |                                          |
| <b>Flammability (solid,gas)</b>                  | Not applicable           | Liquid                                   |
| <b>Explosion Limits</b>                          | No data available        |                                          |
| <b>Vapor Pressure</b>                            | No data available        |                                          |
| <b>Vapor Density</b>                             | No data available        | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>                | 1.0                      |                                          |
| <b>Bulk Density</b>                              | Not applicable           | Liquid                                   |
| <b>Water Solubility</b>                          | Miscible                 |                                          |
| <b>Solubility in other solvents</b>              | No information available |                                          |
| <b>Partition Coefficient (n-octanol/water)</b>   |                          |                                          |
| <b>Component</b>                                 | <b>log Pow</b>           |                                          |
| 1,2-Benzenedicarboxylic acid, monopotassium salt | -3.9                     |                                          |
| <b>Autoignition Temperature</b>                  | No data available        |                                          |
| <b>Decomposition Temperature</b>                 | No data available        |                                          |
| <b>Viscosity</b>                                 | No data available        |                                          |
| <b>Explosive Properties</b>                      | No information available |                                          |
| <b>Oxidizing Properties</b>                      | No information available |                                          |

## SECTION 10. STABILITY AND REACTIVITY

|                                 |                                          |
|---------------------------------|------------------------------------------|
| <b>Stability</b>                | Stable under normal conditions.          |
| <b>Hazardous Reactions</b>      | None under normal processing.            |
| <b>Hazardous Polymerization</b> | Hazardous polymerization does not occur. |
| <b>Conditions to Avoid</b>      | None known.                              |
| <b>Materials to avoid</b>       | No information available.                |

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**Hazardous Decomposition Products** No information available.

## SECTION 11. TOXICOLOGICAL INFORMATION

### Product Information

(a) acute toxicity;  
Toxicology data for the components

| Component                                        | LD50 Oral                 | LD50 Dermal           | LC50 Inhalation |
|--------------------------------------------------|---------------------------|-----------------------|-----------------|
| Water                                            | -                         | -                     | -               |
| 1,2-Benzenedicarboxylic acid, monopotassium salt | LD50 > 3200 mg/kg ( Rat ) | >1000 mg/kg           |                 |
| Potassium chloride                               | LD50 = 2600 mg/kg ( Rat ) |                       |                 |
| Sodium hydroxide                                 | 140 - 340 mg/kg ( Rat )   | 1350 mg/kg ( Rabbit ) |                 |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available  
Target Organs No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available

## SECTION 12. ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component          | Freshwater Fish                                                                               | Water Flea         | Freshwater Algae    | Microtox |
|--------------------|-----------------------------------------------------------------------------------------------|--------------------|---------------------|----------|
| Potassium chloride | Lepomis macrochirus:<br>LC50: 1060 mg/L /96h<br>Pimephales promelas:<br>LC50: 750 - 1020 mg/L | EC50: 825 mg/L/48h | EC50: 2500 mg/L/72h |          |

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|                  |                                                    |  |  |  |
|------------------|----------------------------------------------------|--|--|--|
|                  | /96h                                               |  |  |  |
| Sodium hydroxide | LC50 = 45.4 mg/L, 96h static (Oncorhynchus mykiss) |  |  |  |

**Persistence and Degradability****Persistence**

Miscible with water, Persistence is unlikely, based on information available.

**Bioaccumulative Potential**

Bioaccumulation is unlikely

| Component                                        | log Pow | Bioconcentration factor (BCF) |
|--------------------------------------------------|---------|-------------------------------|
| 1,2-Benzenedicarboxylic acid, monopotassium salt | -3.9    | No data available             |

**Mobility in soil**

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility Highly mobile in soils

**Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

**SECTION 13. DISPOSAL CONSIDERATIONS****Waste from Residues/Unused Products**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

**Contaminated Packaging**

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used.

**SECTION 14. TRANSPORT INFORMATION****Road and Rail Transport**

Not Regulated

**IMDG/IMO**

Not regulated

**IATA**

Not regulated

**Special Precautions for User**

No special precautions required

**SECTION 15. REGULATORY INFORMATION****International Inventories**

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component | The Inventory of Hazardous Chemicals | List of dangerous goods GB 12268 - | TCSI | IECSC | EINECS | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-----------|--------------------------------------|------------------------------------|------|-------|--------|------|-----|-------|------|------|------|------|
|           |                                      |                                    |      |       |        |      |     |       |      |      |      |      |

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|                                                  | (2015<br>Edition) | 2012 |   |   |           |   |   |   |   |   |   |          |
|--------------------------------------------------|-------------------|------|---|---|-----------|---|---|---|---|---|---|----------|
| Water                                            | -                 | -    | X | X | 231-791-2 | X | X | X | X |   | X | KE-35400 |
| 1,2-Benzenedicarboxylic acid, monopotassium salt | -                 | -    | X | X | 212-889-4 | X | X | X | X | X | X | KE-02310 |
| Potassium chloride                               | -                 | -    | X | X | 231-211-8 | X | X | X | X | X | X | KE-29086 |
| Sodium hydroxide                                 | X                 | X    | X | X | 215-185-5 | X | X | X | X | X | X | KE-31487 |

**National Regulations****SECTION 16. OTHER INFORMATION**

**Creation Date** 28-May-2010  
**Revision Date** 04-Apr-2024  
**Revision Summary** Not applicable.

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Legend**

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**PNEC** - Predicted No Effect Concentration

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**Physical hazards**

On basis of test data

**Health Hazards**

Calculation method

**Environmental hazards**

Calculation method

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

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date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**